Bridging Cultures Application – "Talk Link"

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CCT204

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1. Introduction

In this paper, we will present and reflect on our design project on language limbos - “Talk Link”, a user-based social media app. In the following sections, we will present our learning from previous materials, the research study, data collection, data analysis, the identification of the problem and the ultimately design concept. We have acknowledged that the impact of language extends beyond the individual experiences of immigrants and international students, but it is also intricately connected to the broader issue of fairness in the society and overall economy of Canada.

In this project, we employed a qualitative approach, which included an online interview with a female Asian immigrant, Linda, as well as an open-ended questionnaire conducted with 16 anonymous participants. We used an empathy map method for the data analysis and coding. We have coded that the keywords for language barriers are “confidence”, “emotional frustration”, “academic”, “accessibility”.Through the analysis, we gain insights into the personal experiences of those groups and thereby uncover the issues that are commonly present in the Canadian language system. We have identified the most important problem for them is the lack of a pathway of integrating into community and connection with locals, as well as the lack of confidence and anxiety resulting from the psychological stress.

To address this issue, we chose to design a communication-based social media platform. We believe that the best way to provide assistance is by offering pathways to integrate, as “accessibility” is the major key term we identified in our research. We used paper prototyping to make our prototype. We have noticed the integrative nature of new media and the advantages of user-driven content. Therefore, we made our software versatile by incorporating various features such as video, calendar and forums.

We initially conducted user tests within team members, after that, we tested our design with other UTM students and people in our surroundings. They suggested that we should provide more languages to enhance adaptability. They also recommended us to incorporate interactive games to increase engagement.

Ultimately, we reflected on our journey. We recognized the significance of our identity of being international students throughout the entire design process. We also understood how language is an indispensable bridge in our lives for communication, therefore further lay the foundation for our future studies in CCIT.

2. Background and Motivation

The design challenge of this study is that it is hard for some immigrants and international students to integrate and successfully adapt to Canadian culture and life. For these newcomers, a key factor in successful integration is proficiency in the local language, primarily English and French. This is not only essential for effective communication, but also plays an important role in accessing education, employment and a variety of other opportunities. However, there are still many people who cannot adapt to local communication because of the language barrier and do not feel secure or experience belonging.

The reason why this design challenge is important is that Canada is a host country, and the immigrant population is increasing every year. According to the official data of the Canadian government (2023), "there will be 465,000 new permanent residents in 2023." Among this, Canada's emphasis on education attracts a greater number of international students every year, with "a total of 22,682 residents who are international students with study permits in 2021" (Immigration, 2023).

However, some non-native English speakers still need help to fit into society due to language barriers. This is related to the limitation on daily communication, as well as the reinforced immigrant identity due to non-proficient language, which results in an inability to attain the corresponding social status. Through this study, we hope to identify the fundamental challenges that immigrants and students face regarding language difficulties, to improve their situation.

Immigrant populations play a significant role in Canada's economic growth and overall prosperity, as "they often fill the labor gaps that exist in certain industries" (Tiagi, 2015, p.933). However, language barriers not only prevent these individuals from using their talents and expertise in the areas in which they specialize, it also leads to a more vulnerable position which does not commensurate with the status they should rightly have. The "unequal socioeconomic conditions of linguist capital" (Han, 2012, p.137) make immigrants and students can not be accepted as "legitimate speakers". The voices of these groups may be neglected by more powerful groups in society since they have lost conversational legitimacy due to language barriers.

Moreover, "discourse is a cultural tool which reflects one's identification, identity and desire" (Ilieva, 2012). To integrate into society, language learners will proactively change their way of communication and patterns to gain a self-recognized identity. Canada is a culturally diverse country. New immigrants bring a wealth of cultural traditions, languages and art. However, the inability to integrate into the Canadian language system will lead to immigrants and students also struggling to achieve cultural integration. When language

barriers exist, it will influence the knowledge and culture shared and appreciated. This may lead language learners to remain in a perpetual state of “other” and being marginalized.

The study aims to delve into these fundamental challenges that immigrants and international students encounter due to language difficulties, focusing on improving their situation. This involves not only understanding the direct impact of language barriers on daily life and social status but also addressing the broader implications for cultural integration and participation in society.

In conclusion, the challenge of language integration for immigrants and international students in Canada is multifaceted, affecting not only individual communication and social integration but also having broader implications for cultural diversity and economic participation. By identifying and addressing these challenges, the study hopes to facilitate a more inclusive and equitable environment for all newcomers, ensuring that their voices are heard and their potential fully realized in their new home.

3. Related Literature / secondary research

The 2022 Annual Report on Immigration to Canada's Parliament presents a broad picture of the country's immigration landscape, highlighting a record intake of nearly 406,000 permanent residents in 2021, alongside over 415,000 temporary work permit holders and more than 445,000 study permit holders (Immigration, 2023). While these numbers reflect Canada's commitment to a diverse and inclusive immigration system, they also underscore the potential challenges in integrating such a large and diverse group of newcomers into Canadian society.

Delving deeper into these challenges, "Being and Becoming: A New Immigrant in Canada – How Language Matters or Not" by Han (2012) sheds light on the significant difficulties immigrants face due to language barriers. These barriers impact not only communication but also social integration, leading to isolation and affecting mental health. Han (2012) emphasizes the importance of language training programs and community support groups in helping immigrants overcome these challenges. Such initiatives provide crucial opportunities for language skill improvement and social connection, facilitating smoother adaptation to the new environment.

Similarly, "Language Ideology, Agency, and Heritage Language Insecurity across Immigrant Generations" by Teseng (2021) addresses the issue of linguistic insecurity among immigrant generations. The study reveals that heritage language insecurity is particularly acute among second-generation immigrants, with many feeling inadequate in their ancestral language. This sense of inadequacy often leads to language avoidance and struggles with

bilingual identity, exacerbated by community shaming and purist language ideologies (Teseng, 2021).

Additionally, the research article "Effectiveness of a Peer-based Intervention on Loneliness and Social Isolation of Older Chinese Immigrants in Canada: A Randomized Controlled Trial" by Daniel (2020) provides evidence on the effectiveness of peer-based interventions in reducing loneliness and social isolation among older immigrants. This study suggests that such interventions can significantly improve the psychosocial well-being of this demographic.

In summary, while Canada's immigration system is commendably inclusive and diverse, these studies collectively highlight the complex challenges immigrants face, especially regarding language barriers and social integration. They underline the need for targeted interventions, such as language training, community support, and peer-based programs, to ensure that immigrants not only contribute to Canadian society but also thrive within it.

4. Methodology

4.1 Detailed User Group Profile:

The user group for our designed prototype consists of individuals who have immigrated to Canada either accompanying their children or as refugees seeking a new life due to war. They share the commonality of being newcomers to Canada, unfamiliar with the country's environment, language, and lifestyle, yet compelled to integrate into this new environment. Additionally, their economic situation is not particularly optimistic. They require a low-cost solution to help them understand Canada's environment, learn about life in the country, and find a pathway to learn English, the second language. New immigrants and refugees are often less familiar with Canada's environment and life, making them more likely to experience loneliness and face psychological and social challenges such as emotional distress, depression, and anxiety (Lai et al., 2020).

4.2 Research Plan:

We plan to organize and analyze data in the form of an Empathy Map. During our research on the user group, we observed a limited number of existing methods to assist them in integrating into new communities. Regarding language, there is only one government-provided ESL (English as a Second Language) course and online vocabulary-building software. However, such software typically doesn't rapidly enhance users' proficiency in a second language, often requiring extended periods of learning and

application to achieve significant progress. In terms of community integration, there are currently very few available assistance methods. Therefore, our research aims to explore how to create a prototype to help these newly arrived, non-English-speaking immigrants and refugees in Canada.

4.3 Execution:

We conducted data collection through online surveys and video interviews. In the online survey, we ensured ample response information by asking open-ended questions about participants' native languages, challenges posed by language barriers, and methods of learning English, among other factors. The open-ended format provides respondents with more freedom than multiple-choice questions, allowing them to write responses tailored to their situations, and enabling us to gather more accurate and detailed data. We posted the survey on the UTM section of Reddit forums and distributed it to acquaintances who, although not recent immigrants, could respond based on their experiences when they first moved to Canada. We collected a total of 16 responses from the online survey, including six immigrants from other countries, two native Canadians, and eight international students. Fourteen of these responses were deemed valid, excluding the two from native Canadians, as they did not belong to our user group.

Additionally, we conducted an online interview with a local immigrant, Linda, who had moved from China to Canada many years ago. At the beginning of the interview, we assured Linda that we would not collect personal information and that all data would be used for learning purposes only. In the video interview, she shared many challenges she faced when first immigrating to Canada and discussed her initial job choices and how she ultimately overcame these challenges. Linda's interview sessions, limited to 5 minutes each due to her busy schedule, provided valuable information relevant to our research. With Linda's permission, we recorded and transcribed the video interview for further analysis.

5. Findings

5.1 Communication/Language Barrier

According to the data derived from the interview and online questionnaire, a core difficulty for new immigrants and international students would be the language barrier. When they first arrived in Canada, the most challenging thing was to master the local language and communicate with others effectively in order to live and study in this country.

5.2 Confidence in Communication

One important point that is mentioned by both interviewees and participants is confidence in communication. The majority of participants believed that a lack of

self-confidence in their abilities is one of the major obstacles leading to ineffective communication. Among the sixteen participants in the survey, five of them mentioned that "feeling shy" is a common issue faced by immigrants and international students. Some participants are afraid to speak English for fear of making mistakes. They might be concerned that others will make fun of their accents or pronunciation, so they are shy to open their mouths. As a result, they find it difficult to improve their language skills through everyday life practice.

However, many participants still set a high value on the friendliness of the Canadian native community towards non-native speakers. One participant described the communication with locals as an easy and clear path: "Most people are patient and willing to help you if you are trying to speak English. Just go out in the streets (in Canada) and talk to locals". Most of the participants revealed that Canada is a friendly and tolerant country where people are nice and patient to listen to what you say no matter how your English level is. Linda, as our interviewee, states that she overcame her language problem through bravely stepping out of her comfort zone and trying to communicate with her local colleagues. This highlights language learner's dependence on the local community environment and the significance of community support.

5.3 Emotional Frustration and Isolation

One common difficulty observed among participants is the emotional frustration and isolation occurring when they face the situations where they struggle to effectively construct language on their own abilities. Frustrated emotion is a natural side effect of the linguistic challenges, and it can be demonstrated from three perspectives: misunderstanding, inadequacy feeling and independent limitation. Misunderstanding occurs when language is deficient during a conversation, which can result in embarrassment or conflicts, contributing to frustration. The feeling of inadequacy becomes obvious when individuals think they do not live up to their potential and inability to communicate effectively leads them to frustration. Linguistic barriers can restrict independence as individuals need assistance by others instead of handling the task on their own. This is the major source of frustration.

Meanwhile, isolation can be divided into three aspects. First, social isolation happens when individuals who struggle with language find it difficult to engage in social activities and make friends. Secondly, isolation at the workplace or school can emerge if there is lack of effective communication that has an impact on team dynamics, isolates individuals from their colleagues or classmates. Lastly, speaking the local language fluently is the key to understanding and participating in new cultural events or traditions, otherwise, they might

miss out on all the meaningful cultural activities. The occurrence of cultural isolation further isolates individuals from their community, making them feel lonely and separated.

5.4 Academic Performance

It is an undeniable fact that the language barrier has an influence on academic performance. Some international students illustrated that they could not understand what the professor was talking about in class which can result in a lower grade on the academic performance. Target users face comprehension difficulties such as understanding lecture slides and other course materials. Especially when there is a group discussion, they might not be able to participate actively in the discussion due to the linguistic barrier, as they are concerned that they are unable to express themselves clearly and politely. In this case, they will feel isolated from the group and further diminish their motivation to invest in language learning.

5.5 Access to Media

Linda states that her most effective way of learning language is by reading a large number of books. The survey participants also demonstrated a reliance on media in language learning. One participant mentioned that they primarily use Apps as their main pathway of learning, while several others mentioned that “Watching movies and TV shows” and “immersing yourself in the language” are crucial for language learning. While different participants may have varying preferences for the types of media, most of them still suggest that the inputting and outputting of information is crucial for integrating into the language system, as well as integrating into the culture and society. However, this power to access media is never neutral, as middle-class and student groups have more opportunities for media engagement, while other groups may struggle to accumulate these linguistic knowledge capital.

6. Project Identification

We realize that language barriers put immigrants and students in a disadvantaged position. Inadequation in languages itself can restrict their behavior in daily lives, but it can also indirectly marginalize them, leading to a loss of voice and unequal status.

Based on the responses from our interviewees and participants, their perceptions of the causes of language barriers appear to be diverse but also homogenous. Therefore, we have decided to summarize their main pressing problem as follows: Immigrants and students may experience psychological burdens such as shyness and anxiety due to cultural differences, including language habits and culture differences. These burdens can prevent them from building deep connections with the local community. At the same time, non-students and

immigrants with no economic advantages may lack opportunities to use the language appropriately in media and community, thereby missing out the opportunity to quickly integrate into Canadian culture.

Therefore, our problem is the difficulty immigrants and international students in Canada face in linguistically and culturally integration due to language barriers. They find it hard to speak English and French well, hindering their communication, education, and employment. This leads them to feel left out and not having the same chances as others. Our goal is to help these people overcome these language issues and feel more accepted in Canadian society.

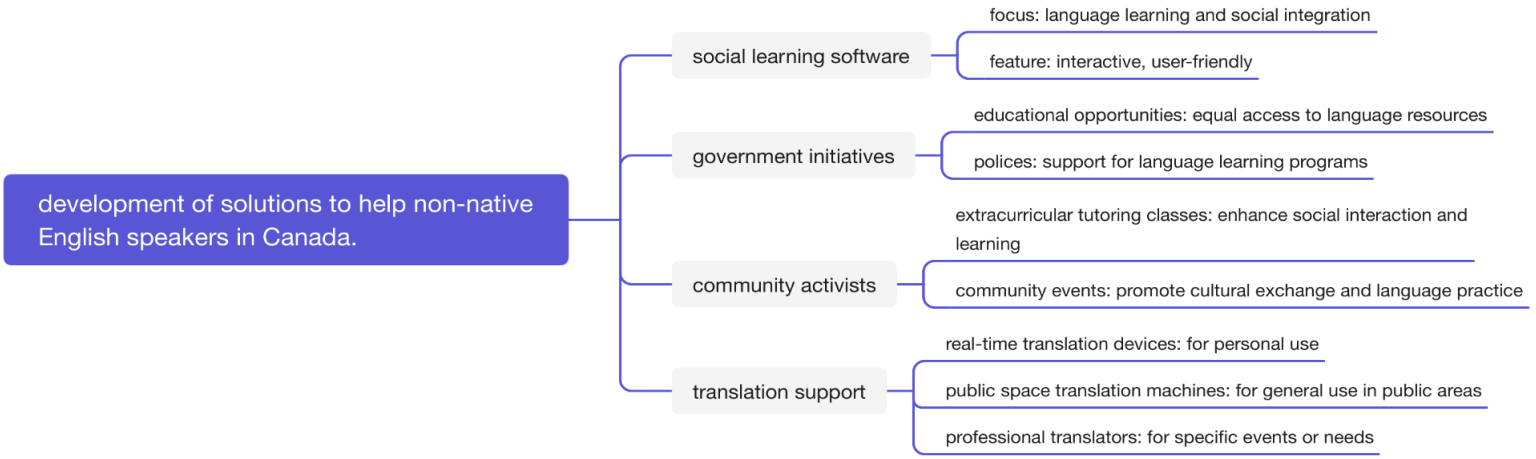
Through pointing out this problem, we hope to find ways in the future to make it easier for these groups to understand and engage with Canadian culture while providing effective media pathways to foster a positive cycle.

7. Ideation

To address the language and cultural integration challenges faced by immigrant and international students in Canada, we began with a brainstorming process. We analyzed survey results to identify core issues and challenges, for example, language barriers, acculturation difficulties, and the common fears and shyness that prevent people from speaking, and in some cases, because of insufficient study or educational background.

Our first idea for a solution centred on the development of new social learning software to help those with English as a second language learn English. The software is designed to promote language learning and social integration. Additionally, we proposed that government departments should offer more educational opportunities, ensuring that everyone arriving in Canada has equal access to language learning resources. Another suggestion was to increase community activities and provide extracurricular tutoring classes, enhancing social interaction and learning opportunities. We also considered distributing real-time translation devices, equipping public spaces with translation machines, or providing professional translators to assist in communication. Our brainstorming process in order to generate as many ideas as possible, recording them without initially considering constraints like funding or time. The central focus was to be user-centric, prioritizing the needs and experiences of the users.

Following the brainstorming phase, we use mind mapping to organize these ideas visually. This method helped us display the content from the brainstorming session in a structured, tree diagram which made it easier to understand and analyze.



Additionally, we incorporated body storming into our process. This involved physically experiencing and empathizing with the challenges faced by non-native English speakers. As most of us are also part of this demographic, it provided a more direct and personal understanding of the issues at hand.

The ideas were then categorized into three main themes: expanding learning options, increasing community activities, and implementing inclusive policies.

7.1 Expanding Learning Options

- Language Learning Apps: These offer the flexibility of learning at one's own pace and are accessible to a wider audience.
- Online Courses: They provide structured learning and can be tailored to different proficiency levels.
- In-Person Lessons: These are highly effective for interactive learning and immediate feedback.

7.2 Increasing Community Activities

- Cultural Events: These events promote cultural exchange and understanding.
- Social Gatherings: Regularly organized gatherings can help build a community network.

7.3 Implementing Inclusive Policies

- Offering translation services in essential public facilities like banks and hospitals, aiding non-native speakers in their daily interactions. These services help reduce the stress of navigating essential services.

Considering these options, we decided to create the English learning app, focusing on language learning with an emphasis on practical and everyday usage. We chose this solution because of the need for a flexible, accessible solution that can reach a broad audience. The

app combines the strengths of language learning apps and online courses, offering a personalized, interactive learning experience. It is designed to adapt to different learning styles and proficiency levels, making it suitable for a wide range of users.

Moreover, the app includes features that allow learners to connect with each other, thus indirectly fostering community engagement. It also provides practical information and language support for navigating daily life in Canada, bridging the gap between language learning and practical application.

In summary, the English learning app represents an approach to language acquisition, it combines the benefits of technology with the human element of learning, ensuring a more inclusive and effective integration process for immigrants and international students in Canada.

8. Prototyping

8.1 Portability:

When designing our prototype, the first consideration for our team was portability at the filtering dimension level. We took into account that our target users are newly arrived immigrants to Canada whose first language is not English. For them, a prototype that helps them learn a new language and integrate into a new environment should be as convenient as possible. This way, they can learn the language through the prototype anytime, without being restricted to a specific location due to the prototype being too cumbersome. Therefore, we plan to design a mobile app as our product, allowing our user base to use our product at any time.

8.2 Functionality:

Next, we also considered functionality during the prototype design. Having a highly integrated app that allows users to fulfill all their language learning and understanding of Canadian culture needs with just one download was a key consideration. In our prototype, we incorporated a forum-like feature, enabling users to post in blocks of their interest using their second language. For example, if I am interested in soccer, I can post about a player's exceptional performance in today's soccer team in the soccer section of the app. Users can mark out the meanings and parts of speech of unfamiliar or complex words in their posts, facilitating learning for others who share the same first language. Users can also comment and like posts under other people's posts, connecting with like-minded individuals. Different-language users can translate unfamiliar or complex words in others' posts into their own language through the software's translation feature, facilitating their learning. Our app also features video chat, allowing users to randomly match with others learning the same

language, communicate with them in this new language, and improve their oral skills. This feature is particularly beneficial as many new immigrants may be hesitant to chat with locals due to fears of having an accent or making mistakes in their second language. However, conversing with someone else learning the same language eliminates such concerns. Our app also includes a calendar function, where users can view national holidays and local community events for the month, complete with explanations and sources for each holiday and event. This provides users with opportunities to delve deeper into the customs of the country and participate in community activities.

8.3 Role Prototypes:

For users, a tool should be convenient and readily available when needed. Therefore, the product we want to design is a mobile app, not any physical product. Smartphones have become essential products that people carry every day, and an app format allows users to access them immediately when needed.

8.4 Look and Feel Prototypes:

When users interact with our product, we aim for a positive user experience where they can quickly find any feature they need. To achieve this, we have placed the functional buttons of our app in icon form at the bottom of the app, making it easy for users to switch to different functions quickly.

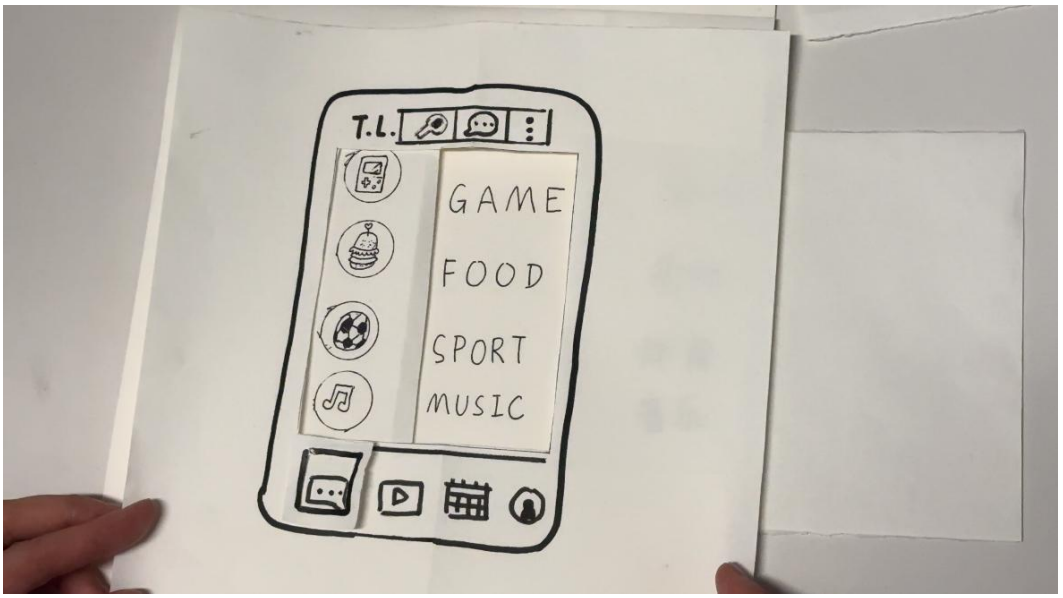
8.5 Prototype implementation

During the prototype development, we employed the paper prototype method to create our design. This approach was chosen for its ability to clearly showcase the overall effectiveness of our prototype, second only to actually developing a functioning app. Below is a link for the video demonstrating the usage of our prototype:
https://youtu.be/ilUVT4QYPiM.

In the initial phase of our prototype implementation, we began by presenting the home page of our design. Located in the top left corner of this page is the icon for our prototype app, Talk Link, featuring the abbreviation "T.L" as a representation of our design. Moving to the top right, three icons are arranged from left to right, signifying the search functionality for users to easily find specific posts, the chat function enabling users to check for replies or private messages (indicated by a small red dot if there are unread messages), and the expandable feature. If users click on the latter icon, a sidebar will appear, offering additional functions such as account security and software language preferences.

In the center of the home page is the main area of our prototype. On the left side of this main area, several circular icons represent different forums. From top to bottom, these

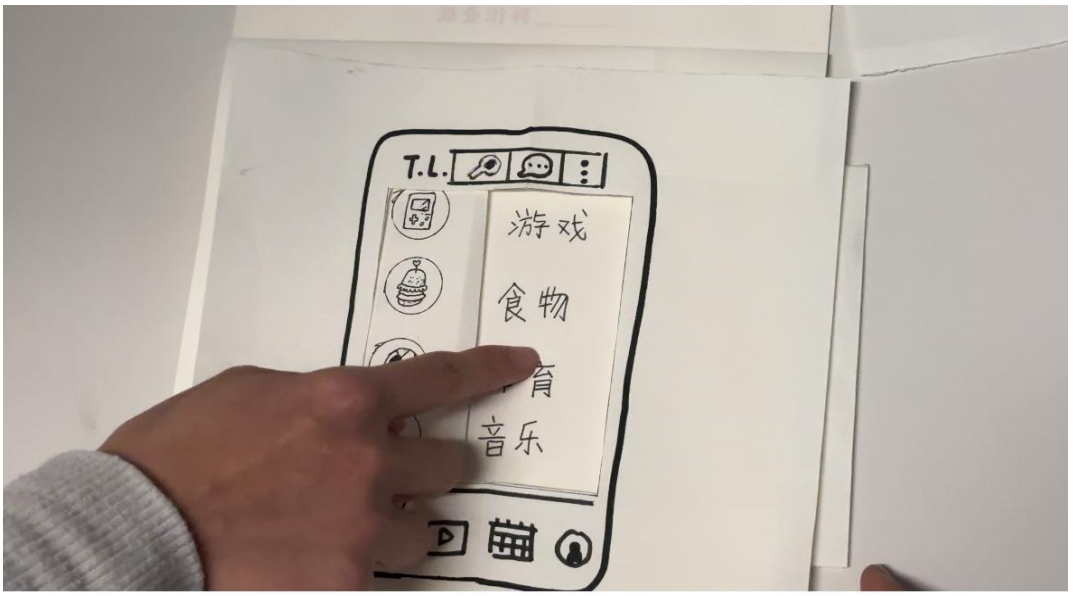
include forums for gaming, culinary, sports, and music. There are many other forums available, but due to the limited size of our prototype, we cannot display the content of all these forums. At the bottom of the main page, there are four icons representing different functions. From left to right, these include forum functionality, video chat functionality, a calendar for checking holidays and local community events, and personal profile functionality.

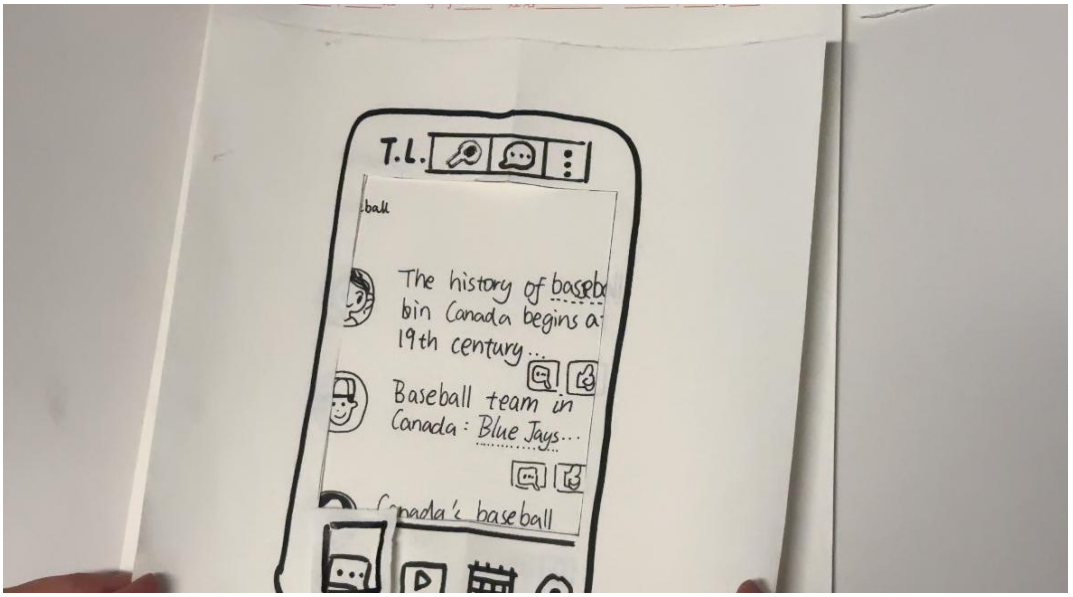


In the forum functionality of our prototype, there is a language-switching feature. Users can easily switch their second language to their native language by sliding the section with text, facilitating the use of our prototype for users who are just starting to learn a second language and may not understand everything initially. This functionality is demonstrated in the image below.



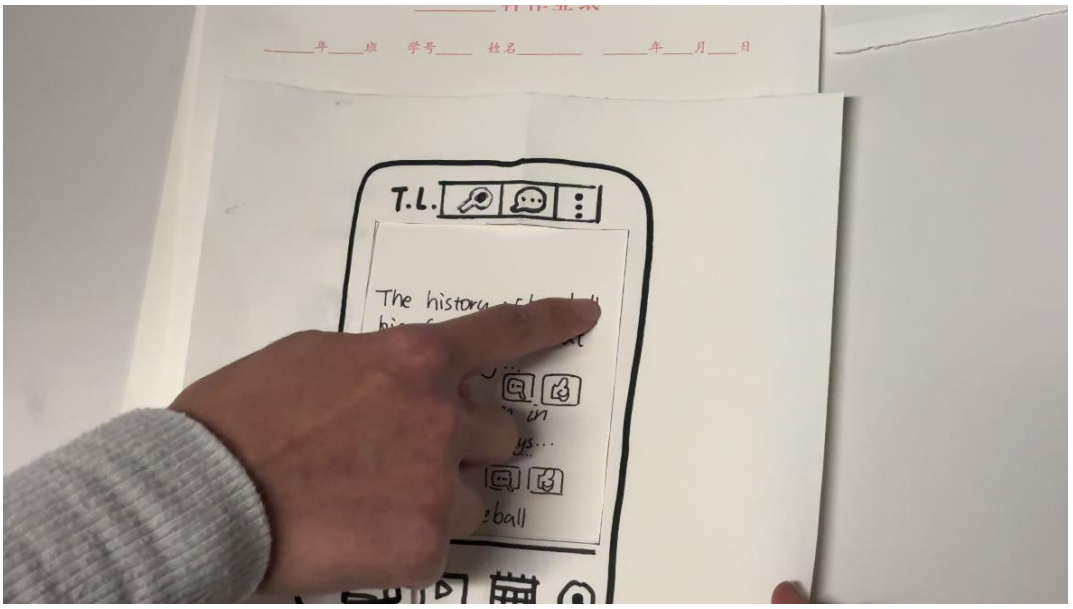
When users click on the text corresponding to any icon on the main page, the app will navigate to the forum page associated with that particular theme. On each post on that forum page, there are functionalities for replies and likes. This allows users to like posts they find interesting, making it easier for them to revisit those posts in the future. Users can also reply to or create posts for threads where they have questions or contributions.

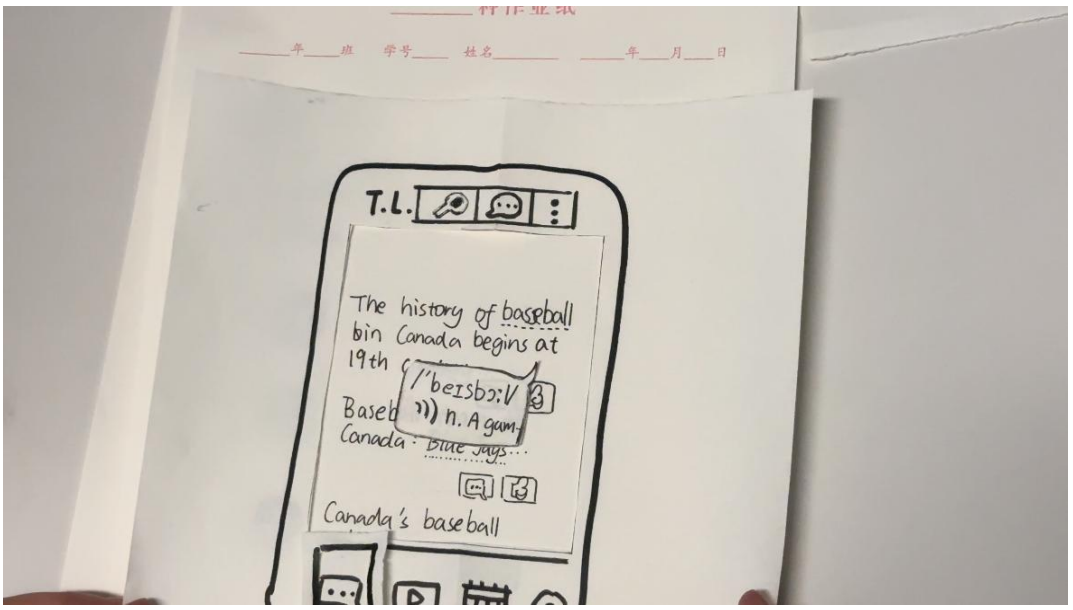




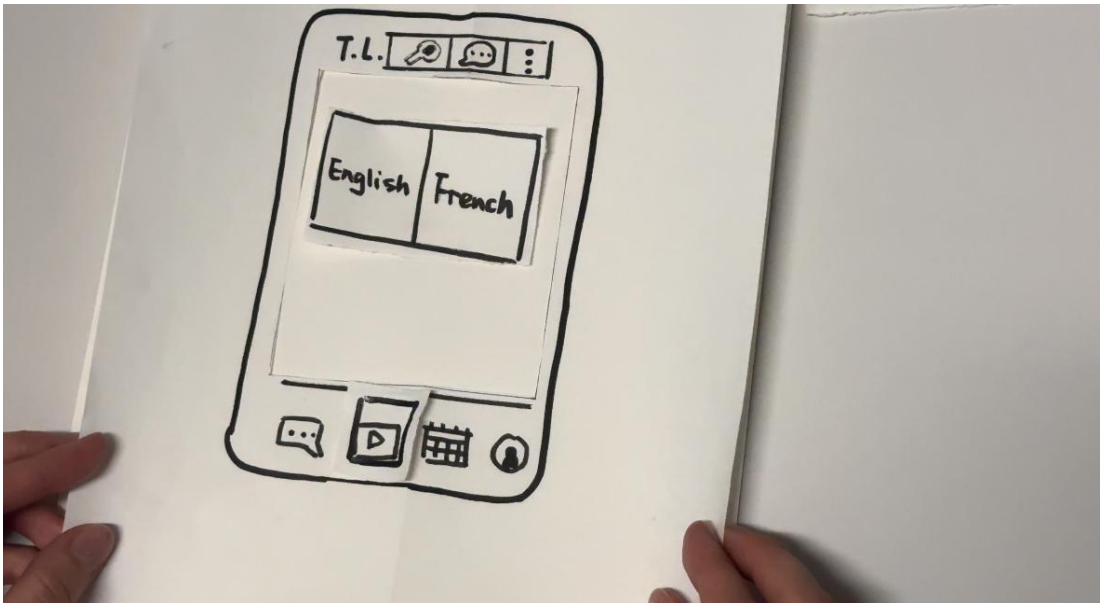
When users click on complex vocabulary explicitly marked by the post's author, they can access information such as pronunciation, part of speech, pronunciation guide, and meaning. This feature facilitates user learning by providing details about vocabulary related

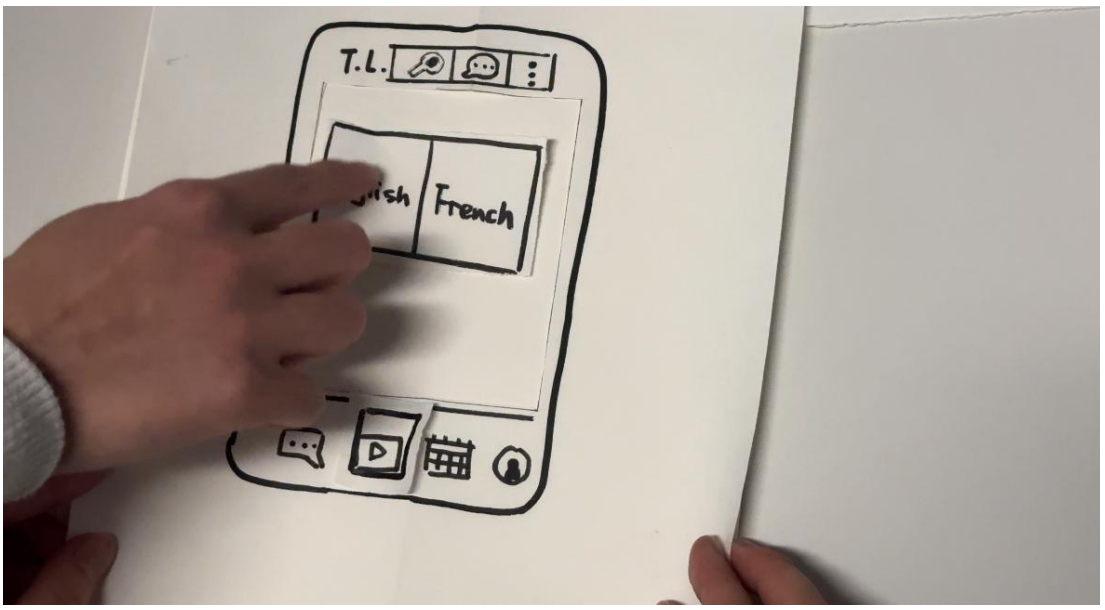
to their interests. It enables users to engage in conversations with others about their hobbies and interests in their second language.



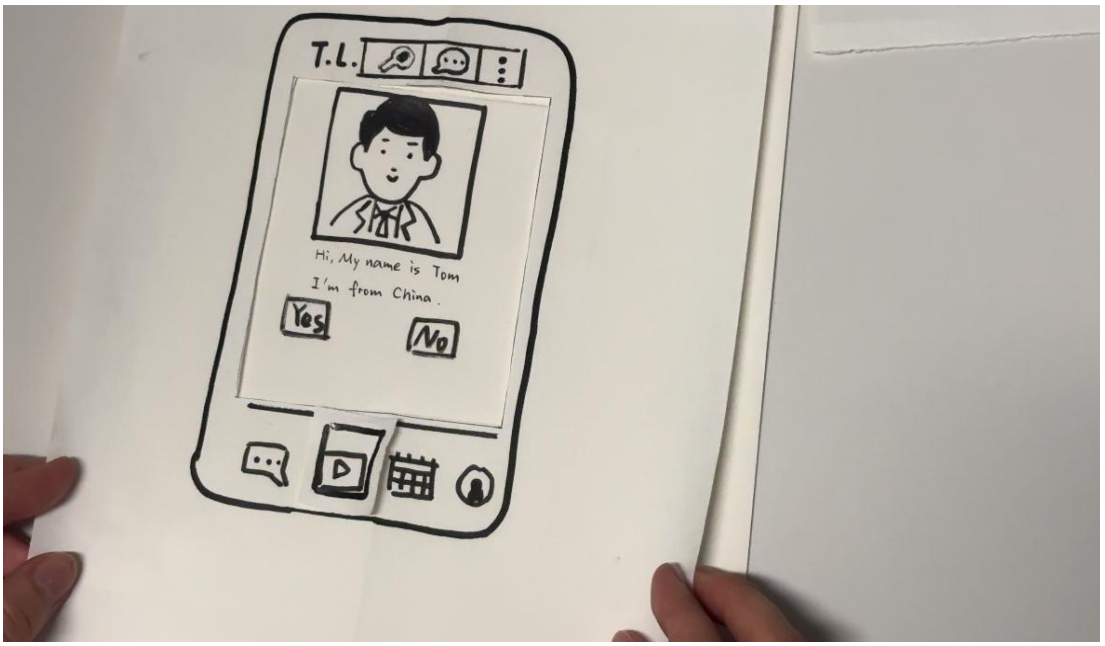


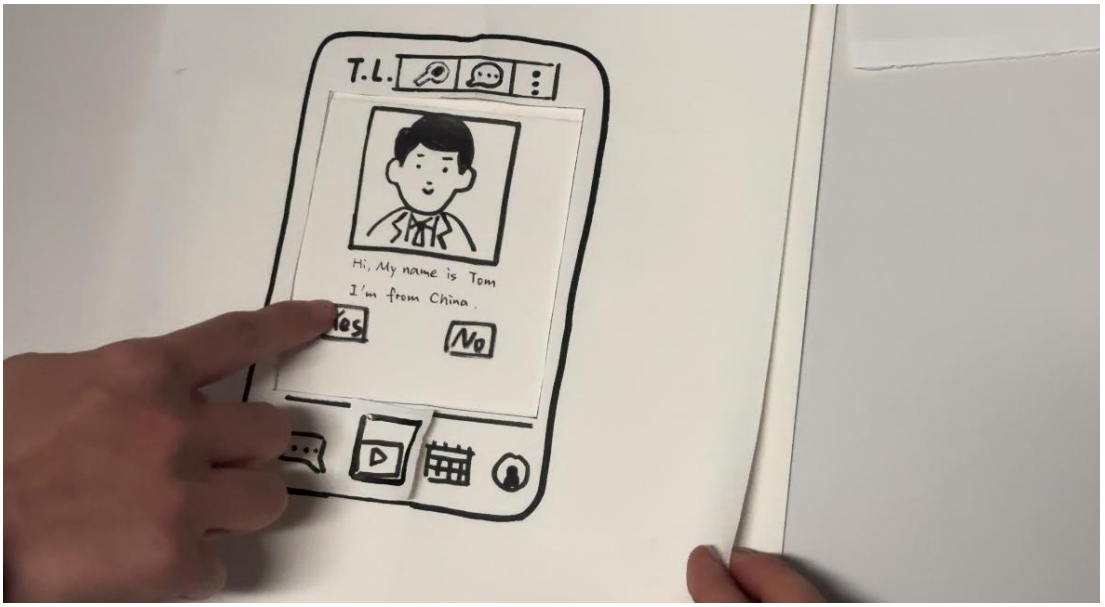
When users click on the video call button at the bottom of the screen, the app switches to the video call page. On this page, users can choose the language they want to use for video calls. For example, if a user wants to learn English, they can click on English, allowing the system to automatically match them with someone who also wants to learn English and is available for a video call in English.



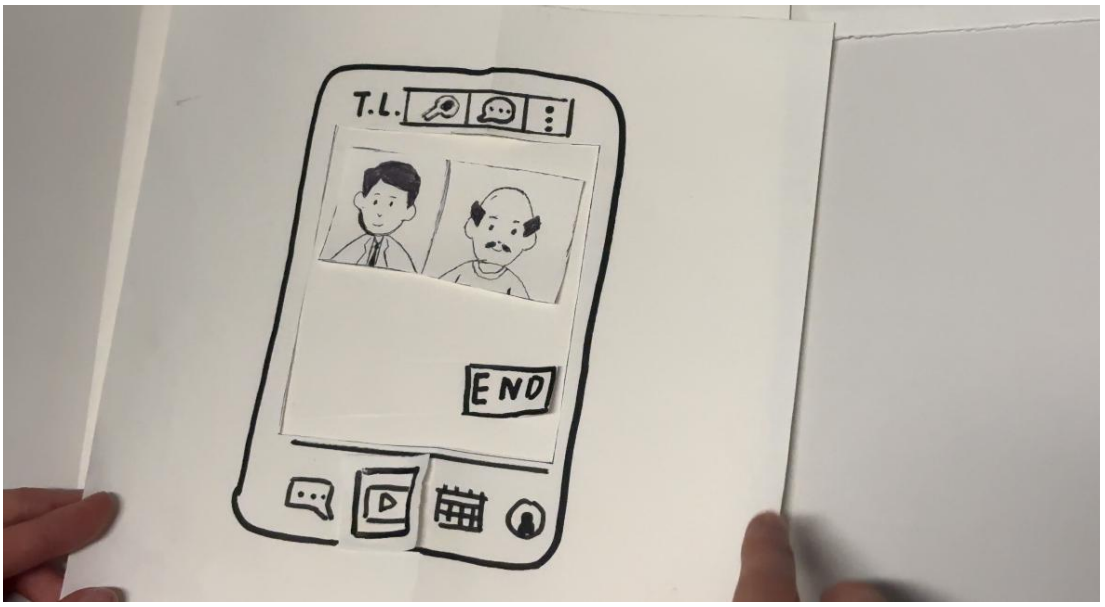


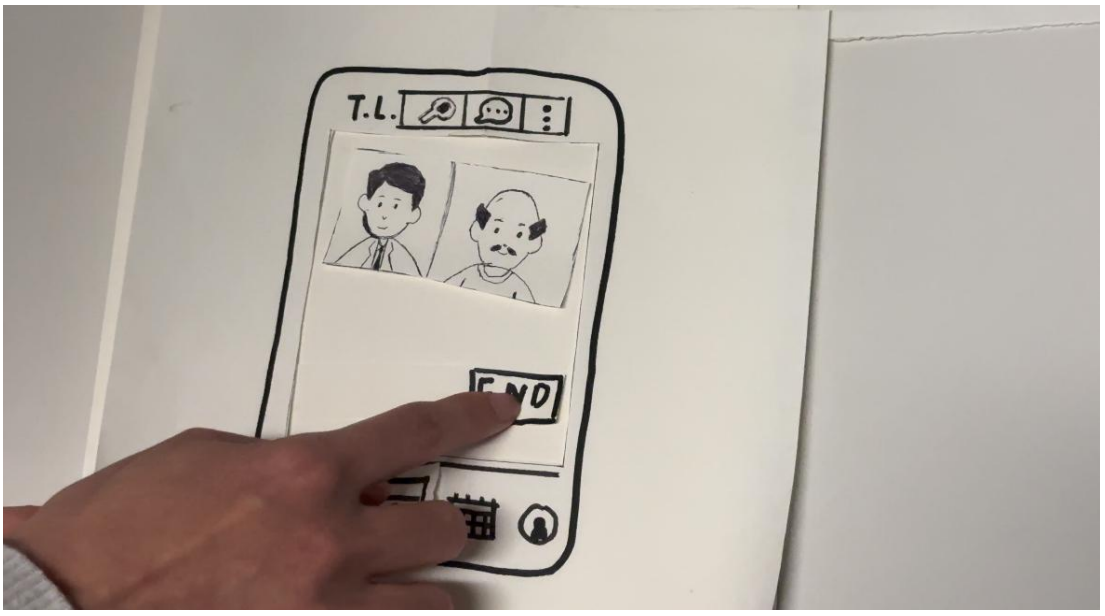
After the user selects the language for video calls, the system proceeds to match the user with a chat partner. When a user is matched with someone, the screen displays the person's real photo along with their self-description. Users can decide whether they want to engage in a conversation with this individual based on the information provided, giving users the power of choice. Additionally, some users may prefer video chatting with individuals of the same gender. In such cases, if a user is matched with someone of the opposite gender, they have the option to decline.



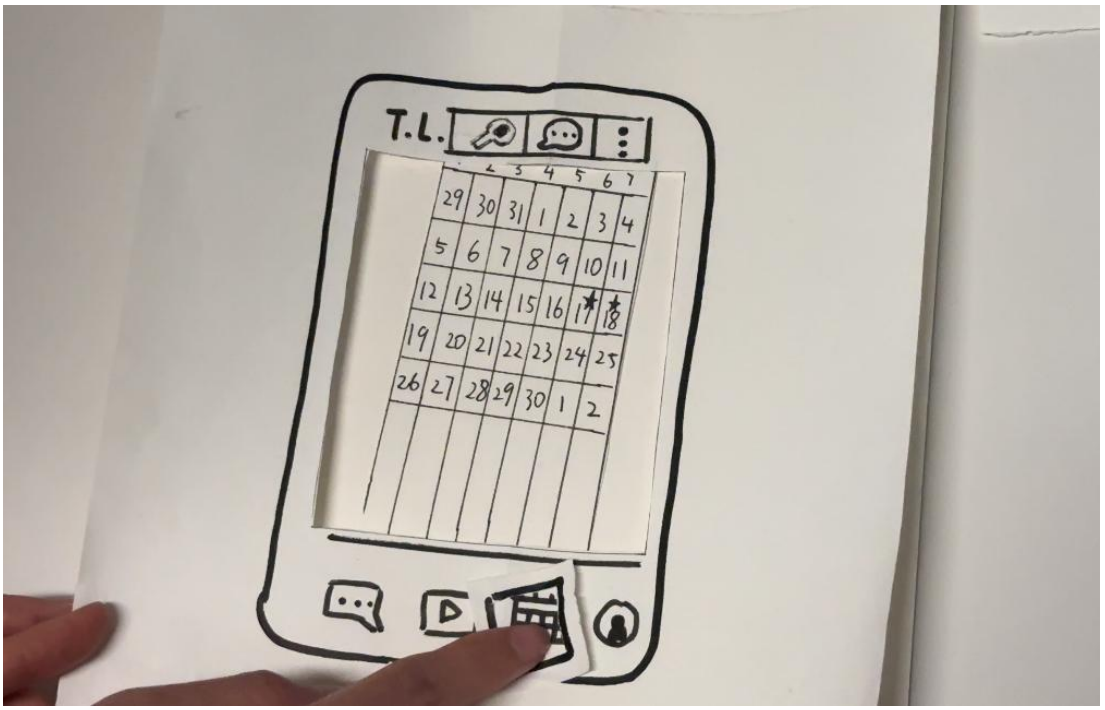


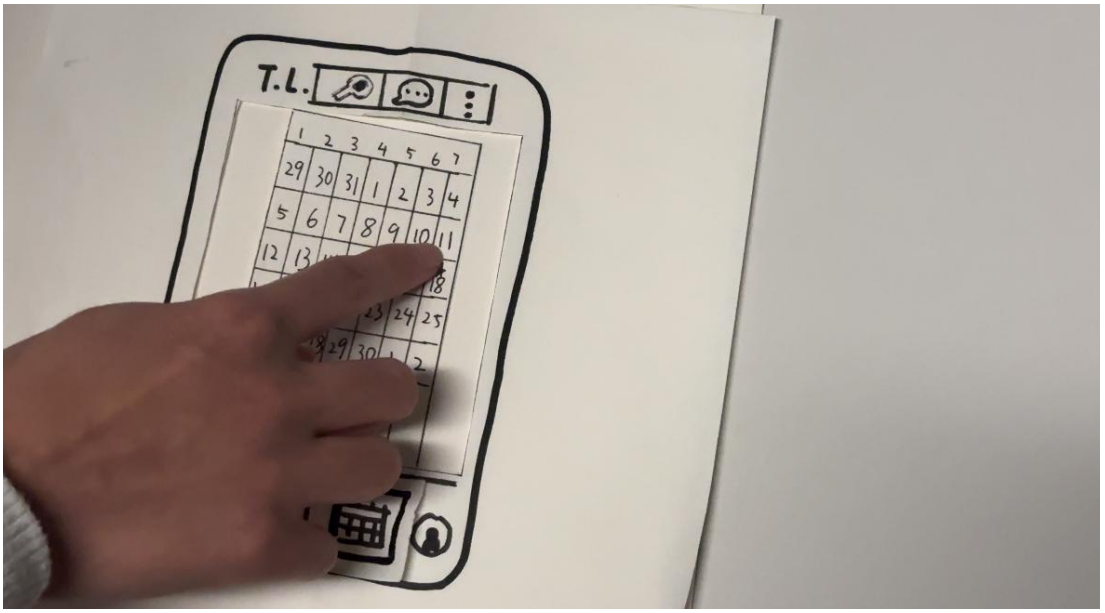
When a user agrees to initiate a video chat, the app transitions to the chat page. When the user feels the conversation is concluding, they can politely say goodbye and press "End." This action concludes a video call.





When users click on the calendar icon at the bottom of the screen, the app navigates to the calendar page. On this page, there is a monthly calendar with a star symbol next to dates that have special holidays or events. This indicates that the local community has special activities or that it is a traditional holiday in that country. Users can click on dates marked with a star to view specific details about the event.



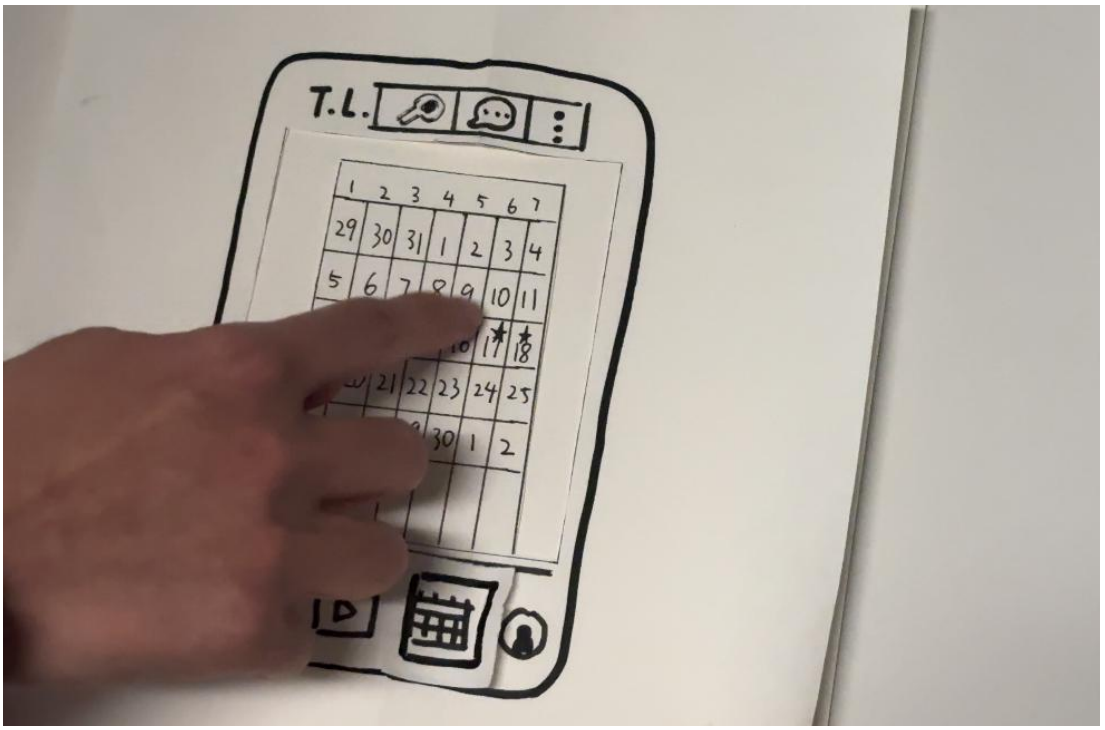


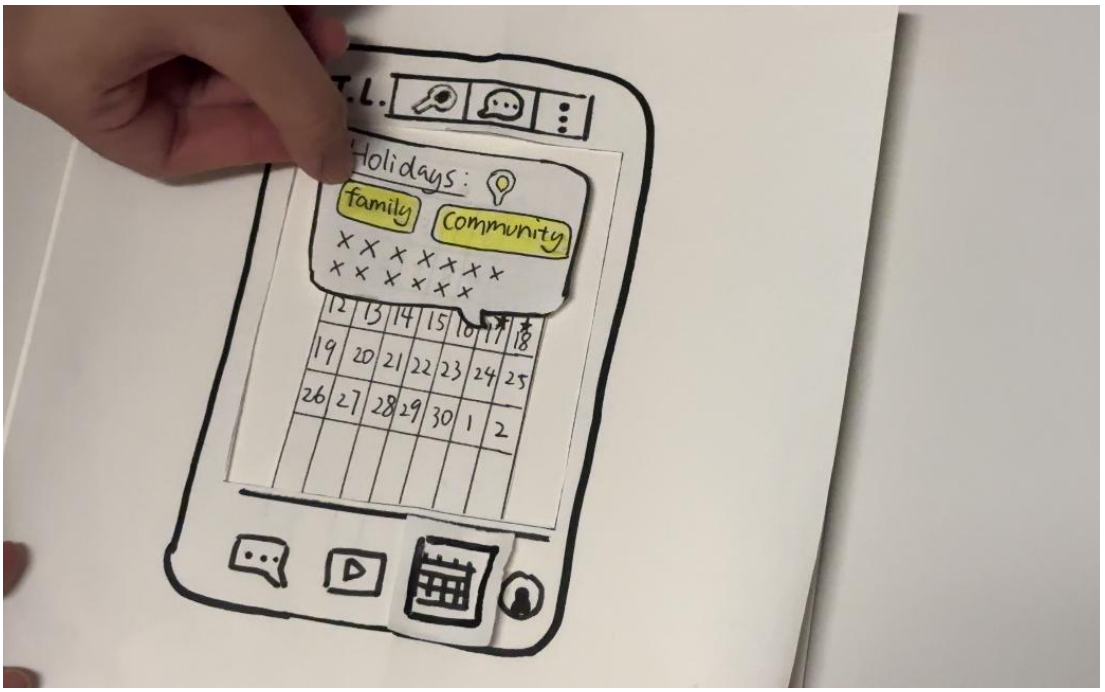
When users click on the video within this interface, the app automatically plays an introductory video about the holiday, presenting the customs in a more vivid format. This approach serves as a dynamic way to introduce the cultural aspects of the holiday to users. Moreover, it is particularly beneficial for individuals with reading challenges, such as those with disabilities, as well as for elderly individuals with visual impairments. Users who prefer textual information can still deepen their understanding of the holiday by reading the

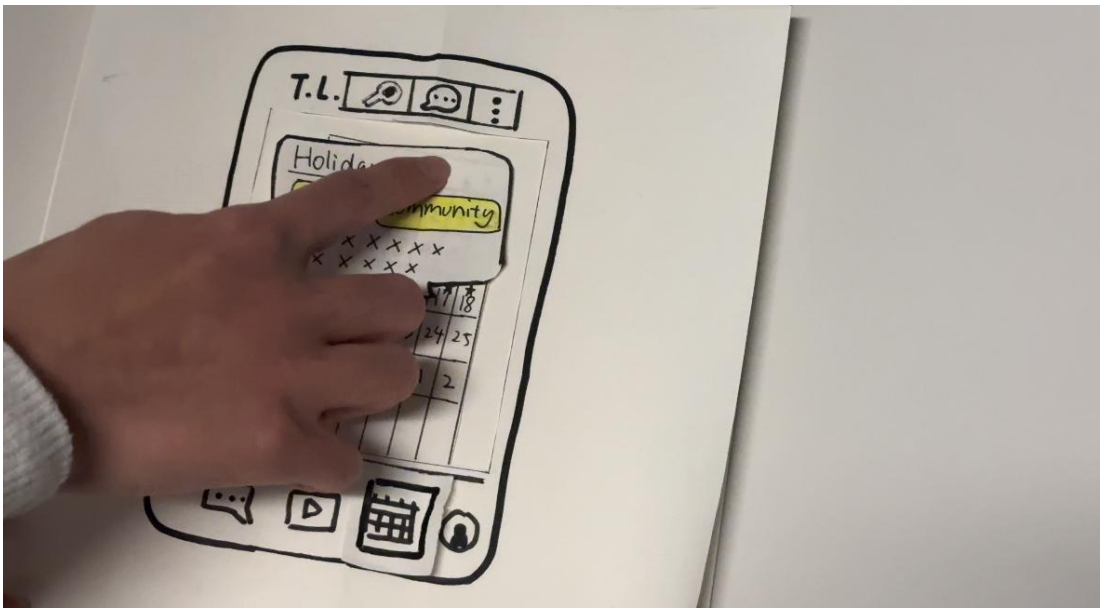
descriptive text provided in the interface. This ensures that users with different preferences can acquire knowledge about the country in a way that suits their preferences.

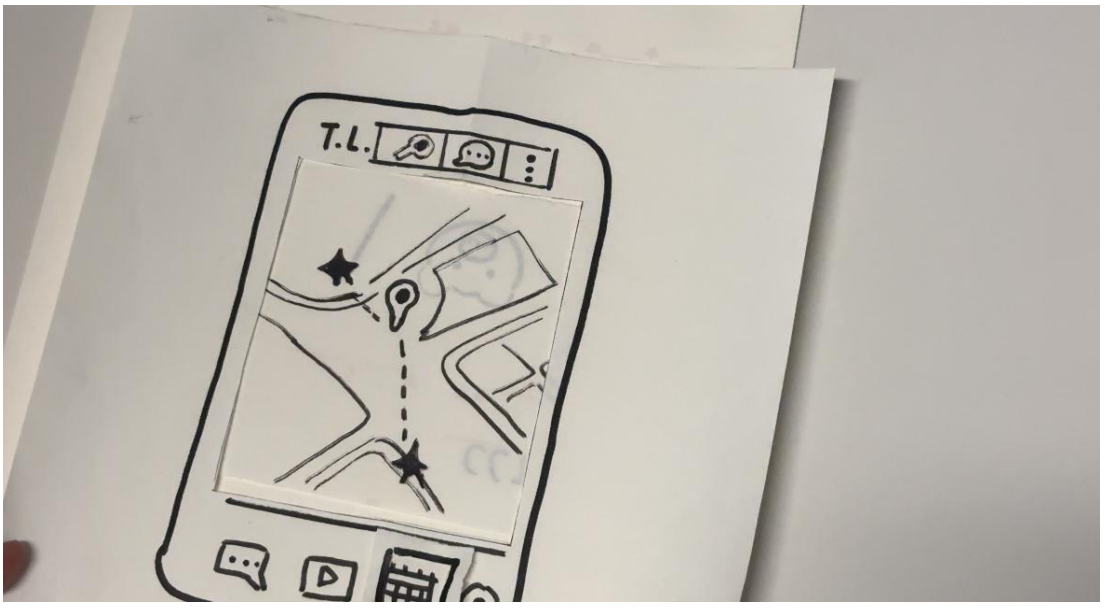


When users swipe right on the screen, the page will return to the calendar. Upon clicking again on a date marked with a star symbol, a prompt will appear on the interface. At this point, users can click on the lightbulb icon, and the page will navigate to a map showing locations where users can find social and language learning venues. This feature assists users in integrating into the local community by guiding them to places where they can socialize and learn English.

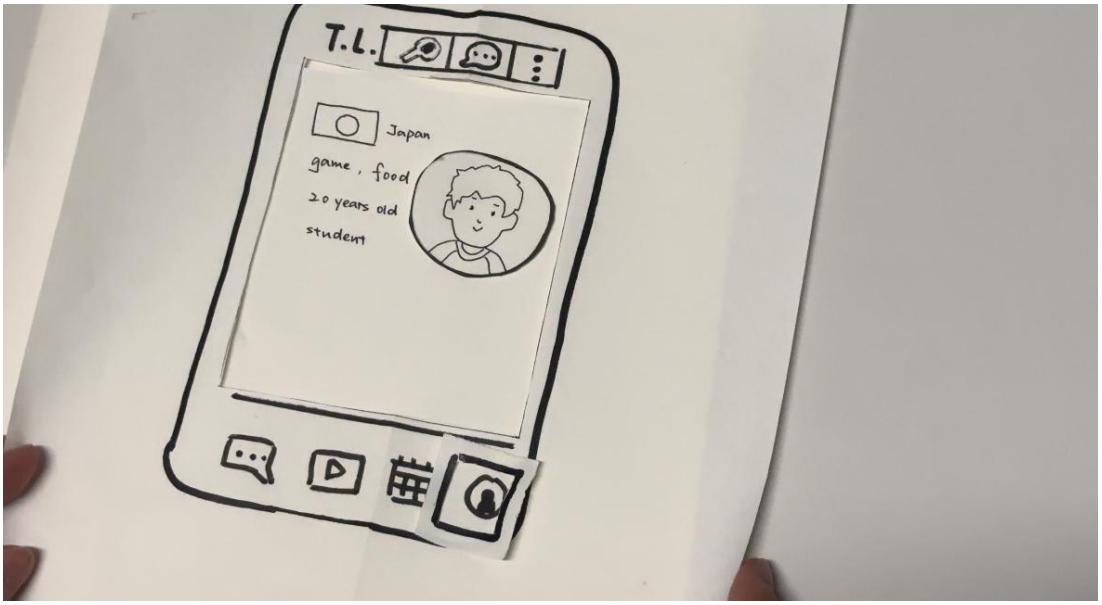



T.L.
Holidays:
family
community
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 1 2


T.L.


T.L.

When users click on the icon in the bottom right corner representing their personal profile, the page will redirect to the user's profile page. The information displayed on this page includes the user's nationality, nickname, interests, age, and role, along with the user's real photo. Users are not allowed to use virtual avatars because, during video chats, it is essential to judge the other user's identity and appearance based on their real photo. To protect user privacy, the nickname does not need to be the real name; users can use a preferred virtual nickname.



9. User Testing

9.1 User Profiles

We conducted two rounds of user testing for the prototype, each involving a different participant. One of the participants is a friend of our group member, currently in her third year at UTM. She is an international student who faced language challenges during the early stages of her study in Canada. Overcoming these difficulties through her own efforts, she walked a challenging path in learning a new language. Due to her personal experience, she expressed a keen interest in our prototype, eager to understand how it could assist users in efficiently learning a new language.

The second participant is a roommate of one of our group members, a Canadian native currently in his second year at UTM. While he does not fall within our primary user demographic, our group aimed to gain insights from a local perspective to identify potential flaws and areas for improvement. Unlike international students or immigrants, local participants may prioritize functionality over interactivity and media richness in software,

focusing on whether the application truly provides an effective language learning channel and methods for integrating into the local community.

9.2 Testing Methodology:

We conducted the user testing in a study room at the library, providing a conducive environment for group discussions, and ensuring privacy and a positive experience for the participants. The testing took the form of interviews. We initially provided a brief introduction to the prototype and allowed participants to explore it without interference. Tasks were assigned to them during the usage, such as finding their personal profiles or entering forums of interest. As researchers, we observed and recorded their reactions, expressions, and the time they spent contemplating each step. Minimal explanations were provided beyond the initial introduction to capture the most authentic responses, simulating the experience of a user using the prototype for the first time.

These user testing sessions were crucial for gathering valuable insights into the usability and potential improvements of our prototype.

9.3 Feedbacks From Users

After the test participants experienced our prototype, we asked for their feedback on the user experience to identify areas for improvement. The participants provided two key points of feedback:

1. **Increase Language Support:** Participants suggested expanding the language support of the app. Currently, our prototype only caters to users whose native language is Chinese. However, this limitation is unfriendly to users of other native languages who are learning a second language to integrate into a new country. The participants proposed enhancing the language functionality of the app to assist individuals from different countries and backgrounds in learning new languages.
2. **Add Interactive Games:** Participants believed that we should introduce more interactive games within the app. They expressed the view that users can better learn a language that is unfamiliar to them through the medium of games. Games provide an entertaining and engaging way to learn, avoiding the dullness associated with traditional vocabulary memorization apps. The addition of interactive games is seen as a means to stimulate users' enthusiasm for language learning while also increasing user engagement.

10. Design Refinement

10.1 Multilingual Expansion

After completing our user testing phase, we received valuable feedback suggesting there are few extra features that need to be added. First of all, the integration of multilingual support. In our original prototype, we only include Chinese and English for the users. In order to meet more users' requirements, we have to extend the App's language support to include more languages that are commonly spoken by immigrants and international students. In this way, it acknowledges cultural diversity, fostering inclusivity and acceptance. Moreover, this expansion catches a wide range of users, ensuring it is more accessible and effective for a diverse user base. By adding more language options, our app will become more inclusive to satisfy varied linguistic needs of immigrants and enhance their experience. It is crucial for an app focused on community integration and support to include this feature, as it directly addresses one of the primary challenges faced by the immigrants: language barriers. It not only improves user satisfaction but also increases the app's reach and impact.

10.2 Interactive Games

Integrating interactive games is an innovative approach to enrich the user experience. By playing these games, users could gain more understanding of a country's language and culture. Cultural quizzes can be added to cover various aspects of different cultures, such as customs, cuisine and history, which encourages users to learn about and appreciate diverse cultures. Besides, role-playing scenarios is an interesting way for users to get familiar with a new environment where they navigate everyday situations in a new culture or language, such as ordering food, attending social events and asking for directions. Last but not least, language learning games should be included where games that teach basic vocabulary and grammar in a fun and engaging way by using visual aids and interactive challenges. Incorporating these elements make the app become more enjoyable and most importantly, reinforces its educational and cultural exchange goals, making it a unique tool for cultural assimilation and language proficiency.

10.3 Virtual Community Spaces

Introducing virtual reality features for immersive cultural experiences and online community gatherings in our app can present a significant enhancement. Through VR, immigrants and international students can virtually explore Canadian landmarks, participate in cultural events and interact with others in a more dynamic environment. This feature enriches a sense of belonging and community among users, improving their integration process. Furthermore, VR can be used for educational workshops, training sessions and stress reduction, which provide a multi-functional platform for different needs. Customizable VR

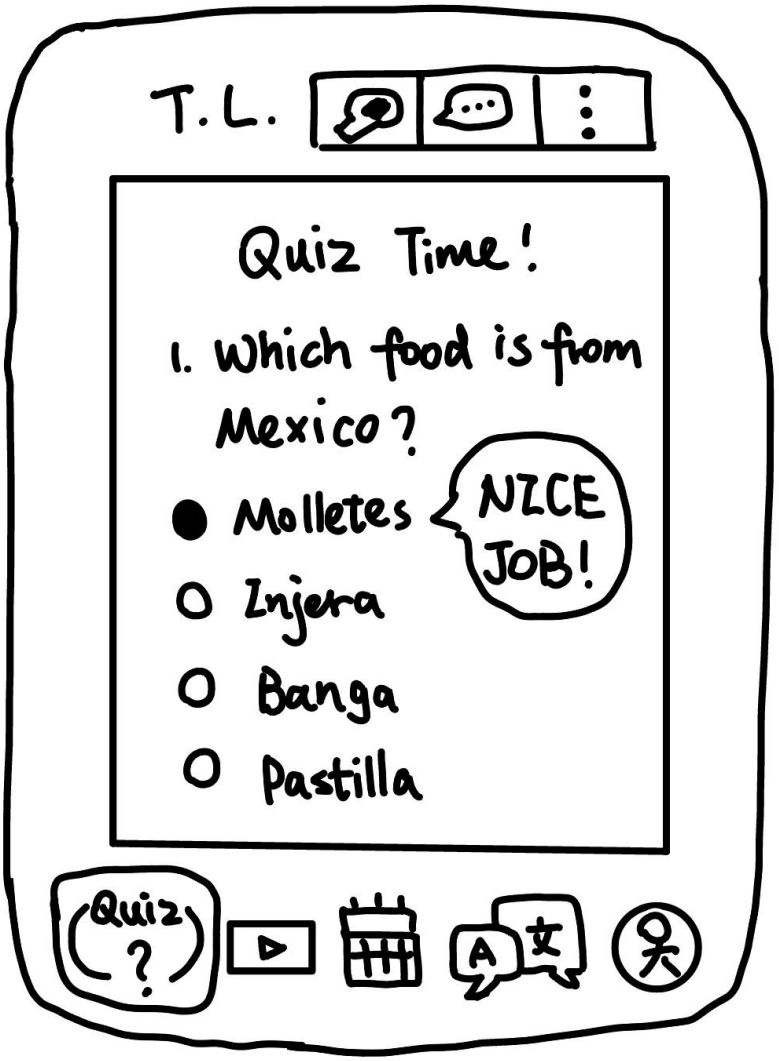
experiences cater to individual interest and requirements, making cultural exploration more accessible and personalized, therefore improving user experience in the app.

10.4 Extra Features Added in Prototype

Figure 10.4.1: Multilingual Support



Figure 10.4.2: Interactive Games



11. Critical Reflection

In the process of the design project, besides the learning and applying of design skills (such as sketching, storyboarding and paper prototyping), we have also gained insights and a deeper understanding on the importance of language. This has increased our resonance with language-related identity, which enhances our connection with immigrants and international student groups.

We mentioned that being international students was one of the reasons that motivated us to initially choose the language limbo as the challenge. An unskilled language not only

represents difficulties in terms of communicating, but also indicates that they are at a disadvantage when it comes to integrating into the local community or advocating for their rights, especially in settings such as banks, courts, or hospitals. At the same time, they may also endure mental stress due to the anxiety of being unable to express or experiencing a sense of being treated differently. As English-learners ourselves, we deeply associate ourselves with these feelings. Through the challenge, we not only identify the pain points for immigrants and other international students, but also make a deeper awareness of the identity of ourselves.

In the research process of the project, we have deepened our understanding of the overall experiences of immigrants and international students in Canada. While we ourselves belong to these groups, the additional findings allow us to contemplate from a more macro perspective. We realized the disparity between the imagined status of immigrants and the reality, as well as the role language plays in these inequalities. Many Chinese immigrants, whom we are most familiar with after the data collection, are predominantly “skilled immigrants”, meaning that they contribute to Canada’s foundational workforce. However, language barriers may impede their opportunities to transition in terms of career.

In the design process of the project, we considered creating a convenient translation software at first. Yet, we abandon this plan and decide to create a social media app. As students of CCIT, we aim to leverage the power of media-related knowledge. Rather than developing a tool which is similar to many existing products in the market and letting immigrants help themselves, we would prefer to see them to truly integrate into the community, with people helping people. This is also a key issue we have identified - language itself serves as a bridge between people, and problems often arise from the dynamics between people, not the language itself.

Last but not least, throughout this design process, we have also realized the importance of teamwork and effective communication. We have come to understand how mutual assistance is crucial in design, or any project, making the entire process inspiring, smooth, and enjoyable for all of us.

12. Conclusions

In this article, we explore challenges that immigrants and international students cannot fit into Canadian society due to the language barriers. Through our research and surveys, we created an English learning app called “Talk Link", specifically tailored for individuals for whom English is not their first language. This application is not just a language learning tool, it also is a bridge to societal integration, designed to instill a sense of

belonging, happiness, and security. After initial usage, users provided valuable feedback which drove us to continually refine and enhance our software, ensuring that it aligns closely with the diverse needs of our audience.

Looking ahead, our focus is on intensifying the promotion of our app to extend its reach to a wider audience. Also, “Talk Link” still requires more promotion to reach a broader user. We believe that “Talk Link” is more than just a language learning platform, it is a catalyst for change. By breaking down language barriers, we are not only helping individuals to communicate effectively but also facilitating their active participation in their communities. We hope that through our app “Talk Link”, users will not only master a new language but also gain the confidence and skills needed to thrive in their new social and cultural settings.

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